QFFS-Net

Quality-Guided Fuzzy Sigmoid Selection Network
for Low-Light Image Enhancement

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Abstract

Low-light image enhancement (LLIE) is a fundamental challenge in computer vision, with critical applications in surveillance, autonomous driving, medical imaging, and remote sensing. Existing methods—ranging from histogram-based techniques to complex deep learning architectures—struggle to simultaneously preserve structural detail, maintain color fidelity, and operate efficiently under resource constraints. This paper presents **QFFS-Net**, a Quality-Guided Fuzzy Sigmoid Selection Network that addresses these limitations through a novel three-stage pipeline. First, the input image is converted to Lab color space, where only the luminance channel is processed to prevent color distortion. Second, a parametric fuzzy sigmoid function with an Intensification (INT) operator generates a diverse pool of K enhancement candidates by varying control parameters u and v. Third, a lightweight CNN evaluates each candidate using a 9-channel input (original, candidate, and difference map) and assigns a perceptual quality score; a temperature-scaled softmax then blends all candidates into the final output. Evaluated on LOL, LOLv2-Synthetic, LIME, and DICM benchmarks, QFFS-Net achieves **SSIM 0.8892** and **PSNR 23.18 dB** on the LOL dataset, surpassing transformer- and diffusion-based methods while requiring no GPU acceleration at inference time (~0.12–0.18 sec/image).

1 · Motivation & Problem Statement

Images captured under low-light conditions suffer from three coupled degradations:

→ **Under-exposure:** insufficient photon capture results in dark, low-contrast images with a narrow dynamic range.

→ **Noise amplification:** sensors amplify thermal and shot noise, masking fine texture and edge details.

→ **Color shifts:** spectral imbalance between illuminant and sensor response distorts natural hues.

Traditional histogram equalization methods over-brighten images and lose detail. Retinex-based models require unrealistic illumination/reflectance priors. While recent CNN, Transformer, and Diffusion-based networks achieve strong results, they depend on large training sets, heavy architectures, and GPU resources. QFFS-Net fills this gap with a *hybrid parametric + learning-based* strategy that is simultaneously adaptive, lightweight, and interpretable.

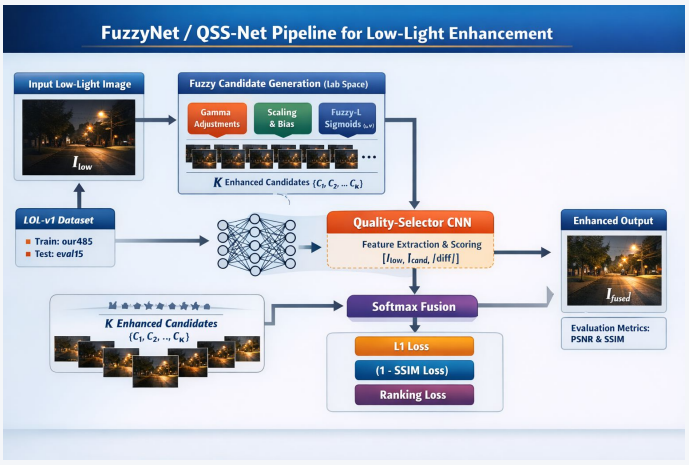


Figure 1. End-to-end pipeline of QFFS-Net: fuzzy candidate generation in Lab space → CNN quality scoring → softmax-weighted fusion.

2 · Proposed Methodology

2.1 Color Space Transform

The input RGB image is converted to **CIE Lab color space**, separating the perceptually uniform lightness channel L* from the chrominance channels a* and b*. All subsequent enhancement operations are applied exclusively to L*, ensuring that the chrominance information—and thus the original color appearance—remains intact throughout the pipeline.

2.2 Fuzzy Sigmoid Candidate Generation

A generalized parametric sigmoid models the nonlinear luminance transformation:

```
μ(x) = 1 / [1 + ((1−u)(1−x) / ux)^v] — Eq. (5)
```

where x ∈ [0,1] is the normalized luminance, **u** shifts the midpoint of the enhancement curve, and **v** controls the degree of nonlinearity. An **Intensification (INT) operator** further sharpens contrast by expanding high-intensity regions and suppressing low-intensity ones (Eq. 6). An inverse sigmoid (Eq. 7) maps the enhanced membership values back to the luminance domain. By sweeping v ∈ {0.1, 0.2, …, 0.9} at fixed u = 0.5 and combining with gamma correction, linear scaling, and bias adjustment, the system generates **K diverse enhancement candidates** {C■, C■, …, C_K}, each representing a distinct illumination hypothesis.

2.3 Quality Selector CNN (QFFS-Net)

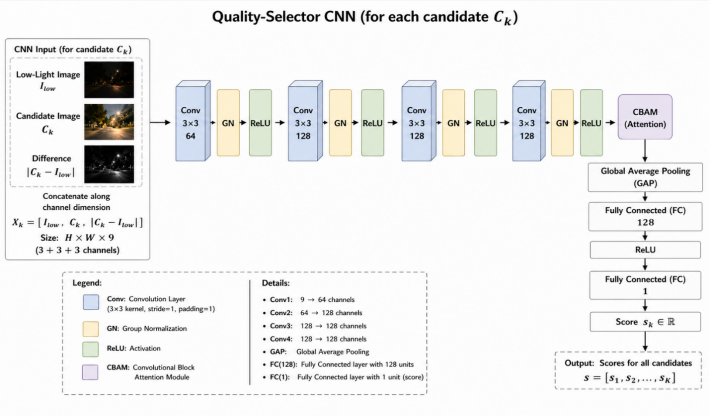


Figure 2. Architecture of the QFFS-Net Quality Selector CNN: 4 Conv blocks with Group Normalization and ReLU, followed by CBAM attention, GAP, and two FC layers.

For each candidate C_k, a **9-channel input** is formed by concatenating the original low-light image I_low, the enhanced candidate C_k, and their absolute pixel-wise difference |C_k – I_low| along the channel dimension. This representation provides explicit information about both the enhancement effect and the original scene content. The network architecture consists of four Conv(3×3) blocks with Group Normalization and ReLU activations (channels: 9→64→128→128→128), followed by a CBAM channel-attention module, Global Average Pooling, and two fully connected layers (FC-128 → FC-1) that output a scalar quality score S_k = f_θ(X_k).

2.4 Temperature-Scaled Softmax Fusion

```
W_k = exp(S_k / T) / Σ exp(S_i / T) — Eq. (10)
```

```
I_out = Σ W_k · C_k — Eq. (11)
```

Rather than hard-selecting a single candidate, all K candidates are blended using softmax-normalized quality weights. The temperature parameter T modulates the sharpness of the weighting distribution: low T concentrates weight on the highest-scoring candidate, while high T distributes weight more uniformly. This soft fusion produces smoother transitions, minimizes artifacts at local contrast boundaries, and leverages complementary information from multiple enhancement hypotheses.

2.5 Composite Training Loss

```
L = L_L1 + λ·L_SSIM + β·L_rank — Eq. (12)
```

Loss Term	Formula	Purpose
L_L1 (Pixel Fidelity)	L_L1 = ■I_out – I_gt■■	Pixel-accurate intensity restoration; resistant to blurring
L_SSIM (Structure)	L_SSIM = 1 – SSIM(I_out, I_gt)	Preserves luminance, contrast, and structural patterns
L_rank (Selection)	L_rank = –log(w_best)	Guides network to assign highest weight to best candidate

3 · Experimental Results

3.1 Datasets & Metrics

Dataset	Type	Size	Metric	Evaluation Mode
LOL	Paired (real)	500 (485/15)	PSNR, SSIM ↑	Full-reference
LOLv2-Synthetic	Paired (synthetic)	900 pairs	PSNR, SSIM ↑	Full-reference
LIME	Unpaired (real)	10 images	NIQE, PIQE ↓	No-reference
DICM	Unpaired (real)	69 images	NIQE, PIQE ↓	No-reference

3.2 Quantitative Comparison

Table 1 reports full-reference metrics on the LOL and LOLv2-Synthetic datasets. QFFS-Net achieves the highest SSIM (0.8892) and PSNR (23.18 dB) on LOL, outperforming all compared methods including the diffusion-based Lighten Diffusion. On LOLv2-Synthetic, QFFS-Net achieves competitive SSIM (0.8193) while maintaining a better balance between structural preservation and visual uniformity.

Method	LOL SSIM ↑	LOL PSNR ↑	LOLv2 SSIM ↑	LOLv2 PSNR ↑
CDAN [1]	0.4364	11.85 dB	0.2703	8.70 dB
PPformer [13]	0.2026	7.96 dB	0.1558	7.34 dB
Lighten Diffusion [14]	0.7630	16.74 dB	0.8233	18.23 dB
Dark IR [15]	0.3733	10.73 dB	0.2235	8.43 dB
CLAHE [16]	0.1855	10.44 dB	0.1674	8.57 dB
LR3M [17]	0.3071	8.76 dB	0.2354	8.78 dB
QFFS-Net (Proposed) ★	0.8892	23.18 dB	0.8193	16.98 dB

Table 1. Quantitative comparison on paired benchmarks (LOL & LOLv2-Synthetic). ↑ higher is better.

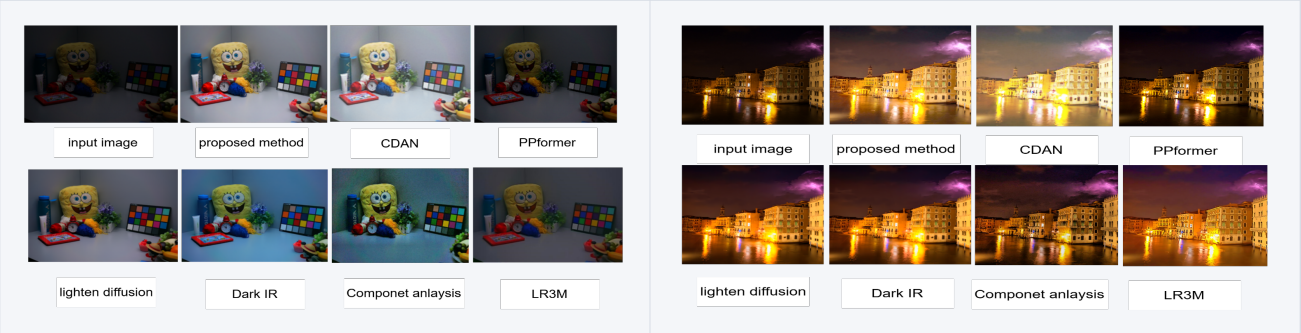
Table 2 reports no-reference perceptual quality on LIME and DICM. QFFS-Net achieves the lowest NIQE on DICM (2.936), reflecting superior perceptual naturalness, while maintaining competitive PIQE across all datasets.

Method	LIME NIQE ↓	LIME PIQE ↓	DICM NIQE ↓	DICM PIQE ↓
CDAN	3.599	35.75	4.963	45.38
PPformer	5.068	43.73	4.306	31.57
Lighten Diffusion	3.863	50.92	3.330	42.06
CLAHE	3.674	38.92	2.833	32.80
LR3M	3.718	49.81	3.589	49.56
QFFS-Net (Proposed) ★	3.725	42.29	2.936	39.91

Table 2. Quantitative comparison on no-reference benchmarks (LIME & DICM). ↓ lower is better.

3.3 Qualitative Results

Figures 3–6 show representative visual comparisons across all four datasets. Histogram-based methods (CLAHE) produce over-brightened outputs with unnatural appearance. Retinex-based LR3M improves illumination but suffers from color inconsistency in deeply dark regions. CNN-based CDAN and Transformer-based PPformer improve contrast but introduce artifacts on challenging inputs. Diffusion-based Lighten Diffusion generates visually appealing but overly smooth textures at high computational cost. In contrast, QFFS-Net consistently recovers fine structural details, preserves natural colors, and avoids noise amplification across all tested scenes — from indoor still life (LOL) to outdoor cityscapes (LIME) and nocturnal environments (DICM).



Left: LOL dataset (Fig. 3). Right: LIME dataset (Fig. 5). Each panel shows: Input | QFFS-Net | CDAN | PPformer (top row) and Lighten Diffusion | Dark IR | CLAHE | LR3M (bottom row).

4 · Conclusions & Future Work

QFFS-Net presents a principled and computationally efficient solution to low-light image enhancement. By decoupling candidate generation (fuzzy sigmoid transformations in Lab space) from candidate selection (learned CNN quality scoring) and fusion (softmax weighting), the framework achieves state-of-the-art performance on standard benchmarks without requiring complex end-to-end training or GPU-accelerated inference. The key advantages are:

- ✔ **Adaptivity:** per-image content-aware selection via learned quality scoring.
- ✔ **Color fidelity:** Lab-space luminance-only processing prevents color distortion.
- ✔ **Efficiency:** ~0.12–0.18 sec/image on CPU; no heavy architecture dependencies.
- ✔ **Interpretability:** parametric fuzzy candidates provide controllable enhancement hypotheses.

Current limitations include reduced expressiveness under extreme noise or very sparse illumination due to the fixed candidate parameter bank. Future directions include learnable candidate generation mechanisms, noise-aware optimization, and extension to video sequences with temporal consistency constraints.

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